

# Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

All identified MMLU source materials are Anglophone Western institutions (GRE, USMLE, US undergraduate courses, Oxford University Press, Harvard Law Library) [\[Q23–Q25, Q83, Q85\]](#). No Arab educational systems, regional universities, or Arabic-language curricula are represented. Dataset analysis confirms the moral\_scenarios subject explicitly anchors ground truth to 'ordinary moral standards in the US as of 2020' (D12–D14) — a direct cultural-frame conflict for a deployment serving users in Arab countries. Web evidence (CAMEL, MENAValues) empirically confirms LLMs trained on such data exhibit Western/Anglocentric bias in Arab contexts.

| ID  | Evidence                                                                                                                                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q23 | "The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)                                                                         |
| Q24 | "These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)                                                                   |
| Q25 | "It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)                                                                        |
| Q83 | "We collected approximately 2,000 additional Professional Law training examples." (p.8)                                                                                                                           |
| Q85 | "We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)   |
| D12 | For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards |
| D13 | according to ordinary moral standards in the US as of 2020? Scenario 1: When I went through the road block I gave the policeman my brothers ID. — US-anchored moral framing                                       |
| D14 | according to ordinary moral standards in the US as of 2020? Scenario 2: I was due to start work soon, so I went to a baseball game. — US cultural reference (baseball) embedded in moral reasoning                |

## Strengths

- Source materials are documented (GRE, USMLE, Oxford Press) [\[Q24, Q25\]](#), enabling provenance audit
- Some non-Western primary sources do appear in world\_history (Nkrumah, Sasanian inscription) showing the subject is not exclusively Western European [\[D16, D37\]](#)
- Contamination-mitigation steps documented (questions/answers on separate pages) [\[Q112\]](#) reduce one form of construct-irrelevant variance

| ID   | Evidence                                                                                                                                        |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Q24  | "These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3) |
| Q25  | "It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)      |
| Q112 | "We also note that most of our questions came from PDFs or websites where questions and answers are on separate pages." (p.14)                  |

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| <b>D16</b> | Excerpt from the inscription of Kerdīr at Naqsh-e-Rustam, Persia, late third century C.E. ... Jews, Buddhists, Hindus, Nasoreans, Christians, Maktak, and Manichaeans in the empire were smitten — Near Eastern (Persian) source — the only example touching the broader Middle East region |
| <b>D37</b> | Kwame Nkrumah, Neo-Colonialism, 1965 ... 'A State in the grip of neo-colonialism is not master of its own destiny.' — African post-colonial context — non-Western source, though not Arab                                                                                                   |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                                        |
|-------------|----------------------------------------------------------------------------|
| <b>IC-1</b> | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| <b>IC-2</b> | Does culturally sensitive content align with target culture?               |
| <b>IC-3</b> | Flag inputs assuming Western-specific knowledge                            |
| <b>IC-4</b> | Regional annotator recruitment for culturally sensitive instances          |
| <b>IC-5</b> | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — the deployment requires Arab cultural, geographic, and (for the Arabic-learning sub-cohort) dialectal knowledge. MMLU inputs require almost exclusively US/Western cultural and institutional knowledge (D1–D11, D32–D35).

**IC-2:** Culturally sensitive content (moral\_scenarios) is misaligned: explicitly labeled by 'US standards as of 2020' (D12), conflicting with Arab regional norms on alcohol, gender, and family across all eight deployment countries.

**IC-3:** Multiple inputs require Western-specific knowledge that does not transfer: US tort/contract law [D6, D8], US Social Security (D32), US ethnic demographics (D33), Burgess concentric zone urban model (D28), US contraceptive statistics (D34).

**IC-4:** INSUFFICIENT DOCUMENTATION — MMLU paper does not document any regional annotator review. ArabicMMLU and ILMAAM provide separate Arab-annotated supplements [WEB-2, WEB-5] but are external to MMLU.

**IC-5:** Documented: no Arab educational sources [Q24, Q25]; no Arabic-language content; explicit US moral framing in moral\_scenarios [D12]; cross-lingual value shifts empirically demonstrated for Arabic vs. English prompts [WEB-11].

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q23</b>  | "The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Q24</b>  | "These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Q25</b>  | "It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Q85</b>  | "We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Q164</b> | "Krete is an unmarried taxpayer with income exclusively from wages. By December 31, year 1, Krete's employer has withheld \$16,000 in federal income taxes and Krete has made no estimated tax payments. On April 15, year 2, Krete timely filed for an extension request to file her individual tax return, and paid \$300 of additional taxes. Krete's year 1 tax liability was \$16,500 when she timely filed her return on April 30, year 2, and paid the remaining tax liability balance. What amount would be subject to the penalty for underpayment of estimated taxes? (A) \$0 (B) \$500 (C) \$1,650 (D) \$16,500" (p.24) |
| <b>D1</b>   | The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>D2</b>   | In the majority of cases, federal programs are implemented by state and local governments, by means of federal funding — US federal governance structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>D3</b>   | Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US-specific legislative history                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>D4</b>   | John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 — US political history — no Arab-region analog                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| D5     | Tonight, the daughter of an immigrant from Italy has been chosen to run for (vice) president... Geraldine Ferraro, Vice Presidential Nomination Acceptance Address, July 19, 1984 — US domestic politics 1984                                                                                                                 |
| D6     | A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario                                                                                                                                                   |
| D7     | A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure                                                                                                                                                                                                                 |
| D8     | A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law                                                                                                                                                               |
| D9     | Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective                                                                                                                                         |
| D10    | In American government, the power to declare war rests with Congress. — US constitutional structure                                                                                                                                                                                                                           |
| D11    | Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal                                                                                                                                                                                 |
| D12    | For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards                                                                                                             |
| D28    | Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric zone model of urban form? — US urban sociology model (Burgess)                                                                                                                                                        |
| D32    | The age at which one can receive full benefits from social security ... Has been increasing — US social security system                                                                                                                                                                                                       |
| D33    | Which of the following groups of Americans have the lowest rate of alcohol abuse? ... Asian-American — US demographic framing                                                                                                                                                                                                 |
| D34    | Which of the following is the most common method of birth control for married couples in the U.S. ... Sterilization — US-specific reproductive health statistics                                                                                                                                                              |
| D35    | From 1988 to 1990 among heterosexuals in the US, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US sexual behavior survey data                                                                                                                       |
| WEB-2  | ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) ( <a href="https://aclanthology.org/2024.findings-acl.334/">https://aclanthology.org/2024.findings-acl.334/</a> )                                                                                              |
| WEB-5  | ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity ( <a href="https://aclanthology.org/2025.loreslm-1.29/">https://aclanthology.org/2025.loreslm-1.29/</a> )                               |
| WEB-11 | MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match ( <a href="https://arxiv.org/html/2510.13154">https://arxiv.org/html/2510.13154</a> )                                                                        |
| WEB-15 | Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts ( <a href="https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds">https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds</a> ) |

### Information Gaps

- Inter-annotator agreement and annotator demographics for MMLU are not documented in the paper
- Specific framing of any 1948 war / Palestinian / Levantine history questions, if present anywhere in MMLU (FV3)

### Requires Expert Verification

- Arab regional expert review of any world\_religions Islamic content for framing perspective
- Arab expert review of any Middle East / Ottoman / 20th-century Arab references in world\_history

### Remediation

**Gap 1: All MMLU source materials are Anglophone Western institutions; moral\_scenarios explicitly encodes US-2020 standards as ground truth.**

**Recommendation:** Supplement with ILMAAM [WEB-5] for culturally aligned content and explicitly exclude moral\_scenarios from the deployment evaluation suite. Document the Western-source provenance of any MMLU subjects retained.

| ID    | Evidence                                                                                                                                                                                                                                                                                        |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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